Partial fractions decompositions

Now that we understand that the Laplace transform is an integral transform that we can use to change a function into a new, predictable form, we want to work toward being able to use this transform to solve differential equations.

But before we start looking at differential equations, let's do a quick review of partial fractions decompositions. Remember that partial fractions decomposition is a technique from calculus we used to rewrite and integrate rational functions.

We'll use these decompositions all the time when we apply the Laplace transform, so it'll be worth our time to go through a quick review of this method here.

Factors and the decomposition

Our first step in any decomposition is to factor the denominator of the rational function as completely as possible. In other words, we want to make sure that none of the factors in our denominator can be broken down any further.

Once the denominator is factored, we need to identify the combination of factors we're dealing with. There are four types of factors we can find: distinct linear factors, repeated linear factors, distinct quadratic factors, and repeated quadratic factors. It's also possible to have a combination of these four types.

© Krista King Math



Function	Factored form	Factor type
$f(x) = \dfrac{2x+1}{x^2-3x+2}$	$f(x) = \dfrac{2x+1}{(x-1)(x-2)}$	Distinct linear
$f(x) = \dfrac{3x-2}{x^2+2x+1}$	$f(x) = \dfrac{3x-2}{(x+1)(x+1)}$	Repeated linear
$f(x) = \dfrac{x+4}{x^4+4x^2+3}$	$f(x) = \dfrac{x+4}{(x^2+3)(x^2+1)}$	Distinct quadratic
$f(x) = \dfrac{x^3-2x^2}{x^4+2x^2+1}$	$f(x) = \dfrac{x^3-2x^2}{(x^2+1)(x^2+1)}$	Repeated quadratic
$f(x) = \dfrac{x^2-2x-5}{x^3-x^2+9x-9}$	$f(x) = \dfrac{x^2-2x-5}{(x-1)(x^2+9)}$	Combination

Linear factors are first degree factors, while quadratic factors are second degree factors. We call factors distinct when they're different from one another, and we call factors repeated when they're the same.

So $(x-1)(x-2)$ is a set of distinct linear factors because each factor is linear, and $(x-1)$ is different than $(x-2)$. And $(x^2+1)(x^2+1)$ is a set of repeated quadratic factors because each factor is quadratic, and (x^2+1) is the same as (x^2+1). It's also possible to have a combination of factors, like $(x-1)(x^2+9)$, which combines one linear factor with one quadratic factor.

For distinct linear factors in our decomposition, we'll use a single constant A for the numerator. For distinct quadratic factors in our decomposition, we'll use a linear factor $Ax+B$ for the numerator, where A and B are constants.

© Krista King Math



Factor type	Factored form	Decomposition
Distinct linear	$f(x) = \frac{2x+1}{(x-1)(x-2)}$	$\frac{A}{x-1} + \frac{B}{x-2}$
Distinct quadratic	$f(x) = \frac{x+4}{(x^2+3)(x^2+1)}$	$\frac{Ax+B}{x^2+3} + \frac{Cx+D}{x^2+1}$
Combination	$f(x) = \frac{x^2-2x-5}{(x-1)(x^2+9)}$	$\frac{A}{x-1} + \frac{Bx+C}{x^2+9}$

Notice how we separated each factor into its own fraction, and only included each distinct factor one time. When factors are repeated, we need to include a fraction for each degree of the factor, up to the highest degree.

For instance, $(x+1)(x+1)$ is a set of repeated linear factors that can be written as $(x+1)^2$. So the degree of this factor is 2, which means we need a fraction for both the first and second degree factors of $(x+1)$.

$$\frac{A}{x+1} + \frac{B}{(x+1)^2}$$

Or let's say that in some significantly more complicated decomposition, we had $x^3(x^2+1)^2$ has a combination set of repeated linear and repeated quadratic factors. The degree of the x factor is 3, which means we need a fraction for the first, second, and third degree factors of x. And the degree of (x^2+1) is 2, which means we need a fraction for both the first and second degree factors of (x^2+1).

$$\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{Dx+E}{x^2+1} + \frac{Fx+G}{(x^2+1)^2}$$

© Krista King Math



As a few more examples, here are the decompositions for the repeated factors from the table above.

Factor type	Factored form	Decomposition
Repeated linear	$f(x) = \dfrac{3x - 2}{(x + 1)(x + 1)}$	$\dfrac{A}{x + 1} + \dfrac{B}{(x + 1)^2}$
Repeated quadratic	$f(x) = \dfrac{x^3 - 2x^2}{(x^2 + 1)(x^2 + 1)}$	$\dfrac{Ax + B}{x^2 + 1} + \dfrac{Cx + D}{(x^2 + 1)^2}$

Let's do an example with distinct linear factors.

Example

Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{2x + 1}{(x - 1)(x - 2)}$$

These are distinct linear factors, so we'll set up the decomposition as

$$\frac{2x + 1}{(x - 1)(x - 2)} = \frac{A}{x - 1} + \frac{B}{x - 2}$$

When the factors are distinct, we'll solve for A by removing the $x - 1$ factor and setting $x = 1$ to find the value of the left side of the decomposition equation.

$$\frac{2x + 1}{x - 2} \to \frac{2(1) + 1}{1 - 2} \to \frac{3}{-1} \to -3$$

© Krista King Math

To solve for B, we'll remove the $x-2$ factor and set $x=2$.

$$\frac{2x+1}{x-1} \rightarrow \frac{2(2)+1}{2-1} \rightarrow \frac{5}{1} \rightarrow 5$$

Plugging $A=-3$ and $B=5$ back into the partial fractions decomposition gives

$$f(x)=\frac{A}{x-1}+\frac{B}{x-2}$$

$$f(x)=-\frac{3}{x-1}+\frac{5}{x-2}$$

$$f(x)=\frac{5}{x-2}-\frac{3}{x-1}$$

Let's look at an example with repeated linear factors.

Example

Rewrite the function as its partial fractions decomposition.

$$f(x)=\frac{3x-2}{(x+1)(x+1)}$$

These are repeated linear factors, so we'll set up the decomposition as

$$\frac{3x-2}{(x+1)(x+1)}=\frac{A}{x+1}+\frac{B}{(x+1)^2}$$

© Krista King Math



When the factors are repeated, we'll multiply both sides of the equation by the denominator from the left side.

$$3x - 2 = A(x + 1) + B$$

$$3x - 2 = Ax + (A + B)$$

Now we'll equate coefficients to see that $A = 3$, and that

$$A + B = -2$$

$$3 + B = -2$$

$$B = -5$$

Plugging $A = 3$ and $B = -5$ back into the partial fractions decomposition gives

$$f(x) = \frac{A}{x+1} + \frac{B}{(x+1)^2}$$

$$f(x) = \frac{3}{x+1} - \frac{5}{(x+1)^2}$$

Now we'll look at an example with distinct quadratic factors.

Example

Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x+4}{(x^2+3)(x^2+1)}$$

© Krista King Math



These are distinct quadratic factors, so we'll set up the decomposition as

$$\frac{x+4}{(x^2+3)(x^2+1)} = \frac{Ax+B}{x^2+3} + \frac{Cx+D}{x^2+1}$$

We'll multiply both sides of the equation by the denominator from the left side.

$$x+4 = (Ax+B)(x^2+1) + (Cx+D)(x^2+3)$$

$$x+4 = Ax^3 + Ax + Bx^2 + B + Cx^3 + 3Cx + Dx^2 + 3D$$

$$x+4 = (A+C)x^3 + (B+D)x^2 + (A+3C)x + (B+3D)$$

Now we'll equate coefficients to see that

$$A+C=0 \qquad B+D=0$$

$$A+3C=1 \qquad B+3D=4$$

Solving both systems gives us $A=-1/2$, $B=-2$, $C=1/2$, and $D=2$. Plugging these back into the partial fractions decomposition gives

$$f(x) = \frac{Ax+B}{x^2+3} + \frac{Cx+D}{x^2+1}$$

$$f(x) = \frac{-\frac{1}{2}x-2}{x^2+3} + \frac{\frac{1}{2}x+2}{x^2+1}$$

$$f(x) = -\frac{1}{2}\left(\frac{x+4}{x^2+3}\right) + \frac{1}{2}\left(\frac{x+4}{x^2+1}\right)$$

© Krista King Math

$$f(x) = \frac{1}{2}\left(\frac{x+4}{x^2+1}\right) - \frac{1}{2}\left(\frac{x+4}{x^2+3}\right)$$

Finally, we'll look at an example with repeated quadratic factors.

Example

Rewrite the function as its partial fractions decomposition.

$$f(x) = \frac{x^3 - 2x^2}{(x^2+1)(x^2+1)}$$

These are repeated quadratic factors, so we'll set up the decomposition as

$$\frac{x^3 - 2x^2}{(x^2+1)(x^2+1)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}$$

We'll multiply both sides of the equation by the denominator from the left side.

$$x^3 - 2x^2 = (Ax+B)(x^2+1) + Cx + D$$

$$x^3 - 2x^2 = Ax^3 + Ax + Bx^2 + B + Cx + D$$

$$x^3 - 2x^2 = Ax^3 + Bx^2 + (A+C)x + (B+D)$$

Now we'll equate coefficients to see that $A = 1$, that $B = -2$, and therefore that

© Krista King Math



$$A + C = 0$$

$$1 + C = 0$$

$$C = -1$$

and

$$B + D = 0$$

$$-2 + D = 0$$

$$D = 2$$

Plugging $A = 1$, $B = -2$, $C = -1$, and $D = 2$ back into the partial fractions decomposition gives

$$f(x) = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}$$

$$f(x) = \frac{x - 2}{x^2 + 1} + \frac{-x + 2}{(x^2 + 1)^2}$$

$$f(x) = \frac{x - 2}{x^2 + 1} - \frac{x - 2}{(x^2 + 1)^2}$$

© Krista King Math